Dual Forest Normalization and Constrained Cancellation for Compressed Unknot Hierarchies

Research continuation for Vladimir Reshetnikov's ProveIt project

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Abstract

The source-certified normal-cut pipeline in ProveIt rejects its finite handlebody test whenever its complementary core contains a three-handle. We remove this restriction by compiling the actual endpoint occurrences of the core two-handles into a rooted dual graph. A spanning tree specifies geometric two-/three-handle cancellations. For the native finite torus-boundary source contract, the graph is always rooted, so all three-handles can be removed in polynomial encoded time, independently of normal-surface multiplicity. The construction retains the exposed port data needed by the existing boundary kernel and has an independent checker.

The choice of dual tree affects subsequent raw one-/two-handle cancellation. We characterize successful choices as spanning trees meeting every stopping set of an unsigned occurrence matrix. This gives complete obstruction-guided search, with a bound parameterized by the number of three-handles and the sizes of the stopping witnesses encountered. If each column containing a unit entry is supported on at most two rows, the problem is polynomial: directed reachability identifies mandatory strongly connected components, and a graphic–partition matroid intersection chooses an independent transversal. Negative certificates in this restricted problem are graphic rank defects. We also prove that each original native column has at most three total occurrences, with its odd entries forming a cycle in the dual graph.

We provide proofs, an additive native implementation, independent certificate replay, finite search implementations, reproducible experiments, and integration instructions. Classical handle cancellation, matroid intersection, and earlier stopping-set results are credited explicitly. These results improve a supplied-cut terminal operator; they do not establish a general quasi-polynomial unknot recognizer.

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1 Research target, source state, and contribution

1.1 The particular obstruction addressed here

The maintained algorithm has several mathematically distinct layers. The complete recognizer can use exact invariant computations and other certifying procedures. The normal-surface research pipeline separately constructs compressed cuts, their parallelity bundles, finite complementary cores, ordered interfaces, and, when a sufficient criterion succeeds, handlebody censuses. A census records every component's genus and binary multiplicity. The latter pipeline supplies potential building blocks for hierarchies, but it is not itself the complete unknot recognizer.

This work fixes one precise gap in that pipeline. At the source revision `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4` [11], both the core certifier and the supplied-cut entry point stop on a nonempty list `handles3`. This is a conservative implementation restriction, not a theorem that the corresponding manifolds are difficult. The retained source charts already contain sufficient information to remove these handles geometrically. Our first contribution is to recover that information, prove the reduction relative to the exposed interfaces, and connect it to the existing independently checked boundary kernel.

The pinned files are in `Topology/UnknotRecognition/fast/fastunknot/`; the principal entry point is `normal_cut_handlebodies.py`. The new branch reuses its already computed parallelity packet. It does not start a second parallelity discovery. The new certificate is dispatched through `normal-cut-handlebodies-v6`. Earlier schemas remain supported. The package includes the patch and a machine-readable source manifest.

1.2 Why this is a continuation of the existing work

We inspected the current implementation, the maintained synthesis, and the ten archives in `docs/incoming` at the pinned revision. Several tempting ideas are already present there and are not claimed as discoveries here. In particular:

- unsigned stopping sets and complete unit-column peeling are established by the incoming stopping-core and saturated-bundle reports;
- cographic persistence uses chamber graphs to retain simultaneously nonseparating surfaces and accounts for causal resets;
- modular Morse certificates verify supplied compressed matchings, with an explicit regular-geometry hypothesis;
- genus budgets, sparse cut grammars, and native ordered interfaces already give several complementary size and attachment bounds.

None of those ten packages implements the native dual two-/three-handle normalization developed below. The combination of dual forest selection with the stopping obstruction also poses a new optimization problem within this development. Its degree-two-column case leads to the polynomial algorithm in Section 4.

Our novelty claims are deliberately local: a source-specific compiled normalization and certificate interface, together with the stated constrained-cancellation characterizations and algorithms. We make no claim that handle cancellation, spanning trees, or matroid intersection are new, and no claim of an exhaustive priority search across all literature.

1.3 Relationship to Lackenby's programme

Lackenby's 2026 preprint [5] presents a hierarchy algorithm for essential boundary patterns and incompressibility. Its Proposition 9.1 bounds iterations by

$$L(g+1)^L, \tag{1}$$

where L bounds hierarchy length and g bounds surface pattern-complexity. Its Section 9 explicitly separates this count from a full running-time analysis. The refined construction gives a linear bound on hierarchy length in Proposition 10.2; the introduction leaves the intended speed-up for further work.

There is also a distinction between two announced exponents. Early Oxford publicity uses $n^{O(\log n)}$ [13]. The author's March 2021 slides give $L = O((\log n)^2)$ and $g = O(n^2)$, resulting in

$$n^{O((\log n)^2)} = 2^{O((\log n)^3)}; \tag{2}$$

the 2024 Tsinghua seminar description repeats this latter bound [4, 9]. Both are quasi-polynomial, but they are not identical. We aim toward the general quasi-polynomial programme without treating a local polynomial operator as a proof of either exponent.

Compressed cutting itself is established prior art. Theorem 9.3 of [3] already gives a polynomial algorithm in triangulation size and logarithmic surface weight, using Agol–Hass–Thurston orbit methods [1]. Theorem 14.4 of [5] gives the uniform-handle version, including bundle type and attachment information. Our contribution concerns an explicit subsequent native operator and its certification. It does not replace or re-prove those general existence theorems.

1.4 Statements proved and statements left open

The main mathematical results are as follows.

1. The native dual graph has $O(t)$ vertices, edges, and incidence work. It is rooted for every admitted source and every admitted normal vector. Its spanning trees give geometric reductions of all core three-handles (Theorem 2.6).

2. This graph depends only on which source edge weights vanish. A tree can therefore be reused throughout such a support stratum, including all positive scalar multiples (Corollary 2.7).

3. For a fixed lower spanning forest, successful dual trees are exactly the graphic bases that intersect every stopping set of the remaining unsigned occurrence matrix (Theorem 3.5).

4. Stopping witnesses give a complete parameterized search; insufficient graphic rank of a family of disjoint witnesses gives a rigorous obstruction (Theorem 3.6 and Proposition 3.8).

5. If every non-inert column has at most two incident rows, tree selection is polynomial and has explicit positive and negative certificates (Theorem 4.4). Every original native column has at most three total occurrences, isolating $(1, 1, 1)$ as the remaining native type (Proposition 4.5); its odd entries form a dual cycle (Proposition 4.6).

The geometric scope is a supplied compressed cut with authenticated charts. The exact search scope fixes the lower zero-/one-handle forest, allows the specified dual-tree reduction, and then allows raw unsigned one-/two-handle cancellation. Failure of this restricted search proves no assertion that the manifold is not a handlebody or that the original knot is nontrivial. Arbitrary handle slides, changes of the lower forest, and hierarchy-wide search remain outside that completeness theorem.

2 Native dual normalization with retained interfaces

2.1 Source contract and size parameters

The native normal-surface validator supplies a connected finite triangulation of a compact orientable three-manifold with one torus boundary. Generalized face pairings and self-identifications are permitted only when the validator's manifold checks pass. The normal vector is nonnegative and integral, satisfies the matching equations and quadrilateral constraints, and belongs to the existing source contract. We do not weaken any of these checks.

Write t for the number of tetrahedra, and B for the largest bit length of a coordinate or local edge weight, with $B \geq 1$. The source constructor `normal_parallelity_geometry.py` partitions the cut into interval-bundle regions and a finite complementary handle core. Its i-handles are denoted H_i. The core has boundedly many local nonparallel regions per source tetrahedron. Crude bounds adequate here are

$$|H_0| \leq 6t, \quad |H_1| \leq 8t, \quad |H_2| \leq 12t, \quad |H_3| \leq 4t. \tag{3}$$

Indeed, a tetrahedron contributes at most two central pieces and four triangle tips; a paired face contributes at most four nonparallel regions; each interior source edge contributes at most its two extreme gaps; and three-handles correspond to interior source vertices. The total number of one-handle crossings in all source two-handle cycles is at most $12t$: each local tetrahedral edge occurs in at most two extreme cycles.

The encoded core is thus small even when a normal coordinate is enormous. Its three-handles retain their source vertex and incident link triangles. Each two-handle retains its source edge, its gap rank, and its full cyclic attaching chart. These records are geometric incidence data; they are strictly stronger than a signed cellular boundary matrix.

2.2 The graph of actual belt-point occurrences

Definition 2.1 (Rooted dual graph)**.** Let K be an authenticated finite three-dimensional handle core. The *dual graph* D has one vertex for each three-handle and a formal root ∂_* for boundary not capped by a three-handle. A two-handle is $D^2 \times I$ and has belt sphere $\{0\} \times \partial I$, consisting of two points. Its edge joins the labels of these two actual belt-point occurrences. A point in a three-handle's attaching sphere receives that vertex label; an uncapped point receives ∂_*.

The root need not be a connected boundary component. It is an algebraic device joining all places at which cancellation can terminate. Edges from the root to itself and edges whose two ends are the same three-handle are loops. A loop has two occurrences, not zero. This distinction prevents a signed boundary coefficient from being mistaken for geometric cancellation.

For native cores, the graph is recovered without tracing an expanded normal surface. Fix an oriented source edge e, with original endpoints u, v and normal intersection weight w_e.

1. If $w_e = 0$, there is a single core gap. Its dual edge joins the labels of u and v.
2. If $w_e > 0$, there are two extreme core gaps. Rank zero joins u to ∂_*, and rank w_e joins ∂_* to v.

An original boundary vertex is assigned ∂_*. The other $w_e - 1$ gaps belong to the parallelity structure and are never enumerated by this step.

Lemma 2.2 (Correct source orientation)**.** *The native source edge identifier is a representative of the signed union-find class of local tetrahedral edges. At that representative the orientation parity is zero. Thus decoding its local edge (a, b) with $a < b$ identifies the original endpoint of rank zero as a, and that of the maximum rank as b, even when the two global endpoint vertices coincide.*

Proof. The parity stored by the signed equivalence structure is the parity between a local edge and its representative. A representative is related to itself with parity zero. The source rank convention reverses ranks precisely when this parity is one. At the representative no reversal occurs, so the claimed identification follows directly from the local edge parameter. The global equality of vertex labels does not identify the two local endpoint germs. □

One must not use an unordered endpoint cache to choose between the two extreme ranks. The implementation decodes the representative itself; the independent checker reconstructs the same geometric fact separately.

Lemma 2.3 (Root reachability)**.** *For every native source and admissible normal vector, every vertex of D is connected to ∂_*.*

Proof. The one-skeleton of the connected source triangulation is connected and contains boundary vertices. Identify all boundary vertices with one root. Boundary edges then become root loops; removing them does not affect root reachability. Keep every interior source edge of zero normal weight. Replace each positive-weight interior edge by two root stubs, one from each of its original endpoints to the root. Replacing an edge in this way preserves root reachability: any path that formerly used that edge can pass through the root instead. The resulting graph, after discarding irrelevant root loops, is exactly the graph just compiled from extreme core gaps. Its nonroot vertices are exactly the interior source vertices, equivalently the core three-handles. □

2.3 Geometric cancellation, including the relative assertion

The following is a source-specific use of the classical handle cancellation theorem: the attaching sphere of a $(k+1)$-handle must meet the belt sphere of a k-handle transversely in one point [6, Theorem 5.4]. Here the belt sphere is zero-dimensional, so the compiled occurrence data give the required geometric intersection directly.

Lemma 2.4 (Rooted dual forest cancellation)**.** *Suppose every component of an authenticated dual graph meets its root. For any spanning tree T after the roots have been identified, there is a sequence of geometric two-/three-handle cancellations deleting precisely the two-handles labeled by T and every three-handle. The surviving two-handles retain their attaching maps to the abstract zero-/one-handle base. No algebraic free reduction or two-handle slide is needed.*

If a specified exposed boundary subcomplex has a collar disjoint from the dual cancellation balls and tubes, the reduction preserves its labeled embedding up to the relative boundary identification.

Proof. Orient the tree toward the root. Choose a vertex adjacent to the already exposed root region. Its parent edge has one belt point in that three-handle's attaching sphere and the other in exposed boundary. The classical cancellation criterion therefore applies. Cancel the pair. Other edges formerly incident with that three-handle now meet exposed boundary. On the dual graph this is precisely contraction of the selected edge into the root. Continue outward along the tree. Each step removes one three-handle; after $|H_3|$ steps the selected tree edges and all three-handles have disappeared.

There is an elementary boundary-ball proof of each step, which also establishes the relative assertion. Write the cancelling two-handle as $h_e = D^2 \times [-1,1]$, with caps $D_e^{\pm}$. Label it so that the three-handle's attaching sphere Σ contains D_e^+ and not D_e^-. Let R contain all other handles. Other three-handle spheres miss both caps, so their attachments can be made before this pair. Then $\Delta = \overline{\Sigma \setminus D_e^+}$ is a disk in ∂R. Together with the attaching annulus A_e of h_e, it forms a disk $E = \Delta \cup A_e$ in ∂R. Meanwhile $h_e \cup_{D_e^+} h_3$ is a three-ball B, and the original manifold is $R \cup_E B$.

Adding a ball along a boundary disk adds only a collar. The collar can be straightened relative to its boundary circle and to every exposed patch in R outside the interior of E.

The move alters the upper attachment boundary but does not introduce a slide of any retained two-handle over another one. Consequently the attaching maps of all retained two-handles to the abstract lower base can be kept as before. This is a statement about the resulting handle description. It does not assert that a homeomorphism fixes every point of the entire old lower submanifold. □

Lemma 2.5 (Native exposed ports)**.** *The exposed core-facing ports of the native parallelity construction can be retained by the reduction in Lemma 2.4, with their incidence, cyclic order, and side labels unchanged.*

Proof. The native constructor places each core-facing port on an exposed region of a zero-handle or a one-handle; its chart records `core_dimension` as zero or one. There are no ports in the interiors of core two- or three-handles. They are actual boundary patches of the complementary core, not patches later covered by a three-handle.

In the boundary-ball proof of Lemma 2.4, every port is therefore contained in R. It misses the interior of E: that interior is an internal gluing region in $R \cup_E B$, not exposed boundary. If a port edge meets ∂E, use the collar straightening fixed on that circle. The relative identification thus fixes all exposed port patches. Iterating this argument retains their product-chart boundary identifications. No port is deleted, reordered, or reattached, so its signed word and its incidence with the unchanged abstract lower charts are retained. □

This argument applies to the authenticated native ports. A proposed future boundary pattern that crosses the cancellation support must supply its own transport. The mere fact that a pattern has a small graph or an Euler characteristic does not meet this requirement.

Theorem 2.6 (Native top-handle normalization)**.** *Given an authenticated native parallelity geometry, all core three-handles can be removed by a certified geometric reduction. Its reduced core and retained exposed interfaces describe the same cut manifold with the same parallelity attachments. The reduction removes exactly $s = |H_3|$ two-handles, preserves Euler characteristic, and needs $O(t)$ graph records.*

Graph compilation and a breadth-first tree use $O(t)$ arithmetic/label operations. Independent union-find verification uses $O(t\alpha(t))$ label operations. The arithmetic involves B-bit weights and $O(\log(t+1))$-bit labels, so a deterministic array implementation has polynomial bit cost, for example

$$O\big(tB + t\alpha(t)\log(t+1)\big)$$

in a logarithmic-cost random-access model, after source authentication, for the canonical bounded-size certificate. The delivered producer uses arrays and breadth-first search; replay uses separate arrays and union-find. A multi-tape Turing implementation also has polynomial bit cost, without asserting this exact random-access bound. For arbitrary malformed input, verification time additionally includes its actual encoded length; a huge invalid identifier must be read before it can be rejected.

Proof. Apply Lemma 2.3, construct a tree, and use Lemmas 2.4 and 2.5. A tree on $s+1$ vertices has s edges. Removing equal numbers of two- and three-handles leaves the alternating handle count unchanged. The size bounds (3) and the local edge-weight formula give the operation bounds. The independent checker needs only to reconstruct the selected edge endpoints, reject cycles, and verify that all s vertices reach the root. The list of selected original two-handle identifiers has $O(s\log(t+1))$ bits; all retained charts can be shared or copied from the authenticated source. □

Corollary 2.7 (Support-stable normalization)**.** *For a fixed source triangulation, the dual graph, using semantic labels* $(e, zero)$*,* $(e, left)$*, and* $(e, right)$*, depends only on the set of edges with zero normal weight. A selected dual tree is valid throughout that stratum. In particular it is valid for every positive integer multiple of a supplied normal vector.*

Proof. The two cases in the graph compiler use only $w_e = 0$ or $w_e > 0$. The endpoint labels come from the source vertices and remain fixed. Multiplication by a positive integer preserves every zero test. Maximum ranks are updated to the current w_e when actual chart identifiers are reconstructed. □

Proposition 2.8 (Optimal initial deletion cost)**.** *Assign a nonnegative cost* c_e *to every native two-handle, such as its attaching-cycle length or its raw occurrence count after a fixed lower forest. A maximum-weight spanning tree minimizes the total cost of the retained two-handles among all dual-tree normalizations.*

Proof. For a tree T, the retained cost is $\sum_{e\in E(D)} c_e - \sum_{e\in T} c_e$. The first summand is constant. Thus minimizing the retained cost is exactly the maximum spanning-tree problem. □

This objective is inexpensive but does not guarantee subsequent raw cancellability. The next sections address that stricter question.

3 Choosing the dual forest using raw cancellation obstructions

3.1 The restricted certificate class

Fix a zero-/one-handle spanning forest F in the core. Contracting F leaves one zero-handle in each component and some one-handles. Every two-handle has a geometric attaching cycle through their belt disks. Let

$$A = (a_{rc}) \in \mathbb{N}^{R\times C}$$

be its unsigned occurrence matrix: r labels a two-handle and c a surviving one-handle. Signs and cancellations in the fundamental group are not used. Multiple geometric crossings contribute multiple occurrences. One may saturate each count at two because only zero, one, and at least two matter to the predicates below.

The same row set R labels the edges of the rooted dual graph D. When interface relations are included, give each row ineligible for top cancellation an artificial root loop. Such a row cannot be selected in a tree; it still participates fully in every stopping test. Choosing a dual spanning tree $T \subseteq R$ deletes its rows, by Section 2. We ask whether all remaining rows can be removed by raw geometric one-/two-handle cancellation. A raw pivot (r, c) requires $a_{rc} = 1$ and has the unsigned update

$$a'_{ih} = a_{ih} + a_{ic}a_{rh} \qquad (i \neq r,\ h \neq c), \tag{4}$$

followed by deletion of r and c. This counts the crossings introduced by pushing the other attaching bands through the cancelling pair. It is the conservative raw update used by the existing geometric certifier. Saturating after each update preserves every future zero/one test.

Definition 3.1 (Forest-compatible raw completion)**.** The pair (D, A) is *completable* if some spanning tree T of D leaves $A[R \setminus T, C]$ completely eliminable by pivots (4). The lower forest F remains fixed throughout this definition.

This problem deliberately excludes normal isotopies that change occurrence counts, arbitrary handle slides, and replacement of the lower forest. Its positive answers are useful geometric certificates. Its negative answers have only the scope of this definition.

3.2 Stopping sets and the exact raw criterion

We recall the stopping-set theorem of the incoming stopping-core work [8, 7], including a proof because its invariant is essential to the new search.

Definition 3.2 (Stopping set)**.** A nonempty set $S \subseteq R$ is a stopping set when

$$\sum_{r \in S} a_{rc} \neq 1 \quad \text{for every column } c. \tag{5}$$

Equivalently every column has zero or at least two occurrences in S. A *unit-column peel* removes the unique row incident with a column whose total is one. The column can be removed as well: it has no other occurrences, so this pivot creates no fill.

Lemma 3.3 (Canonical stopping residual)**.** *Exhaustive unit-column peeling leaves the same row set regardless of tie-breaking. This set is the union of all stopping sets. In particular a row restriction is peelable exactly when it contains no stopping set, and peelability is hereditary under deleting rows.*

Proof. The union of stopping sets is stopping. If a column had total one on the union, its unique contributing row would belong to a constituent stopping set on which that column also had total one, a contradiction.

A row in a stopping set cannot be peeled while the stopping set is intact, because every column through that row has at least two occurrences within the set. Thus every stopping set survives every sequence of peels. At termination, if rows remain, their set is itself stopping. It therefore equals the union of all stopping sets and is independent of the order. Finally a stopping subset of a smaller row restriction would also be a stopping subset of the larger restriction, proving heredity. □

Lemma 3.4 (Completeness for raw pivots)**.** *All rows can be removed by arbitrary raw pivots* (4) *if and only if unit-column peeling removes all rows.*

Proof. Peeling is a particular raw pivot sequence. For the converse, suppose S is stopping and perform a raw pivot (r, c). If $r \notin S$, the total $m = \sum_{i \in S} a_{ic}$ is zero or at least two. In every remaining column h, the new total is

$$\sum_{i \in S} a_{ih} + m a_{rh}.$$

It is zero or at least two: a formerly zero total either stays zero or increases by at least two, and a positive old total was already at least two. Thus S remains stopping after removal of c.

If $r \in S$, then $m \geq 2$ because $a_{rc} = 1$. Put $S' = S \setminus \{r\}$. The set S' is nonempty; otherwise S would already have total one in column c. In a remaining column its total is

$$\sum_{i \in S'} a_{ih} + (m-1) a_{rh} \;\geq\; \sum_{i \in S} a_{ih}.$$

An old positive total was at least two, and an old zero total has all its summands zero. Hence S' is stopping. Inductively a nonempty stopping set survives every pivot sequence, preventing complete elimination. Lemma 3.3 now proves the converse. □

The proof explains why signed exponent sums are inadequate. A curve with raw word $xx^{-1}x$ crosses the belt disk three times, even though its group word freely reduces to x. Treating these as one crossing silently changes the permitted geometric move class.

3.3 Spanning trees as transversals of obstructions

Theorem 3.5 (Forest–stopping-set equivalence)**.** *Assume D is connected after boundary roots are identified. A spanning tree T is a successful forest-compatible raw completion if and only if*

$$T \cap S \neq \varnothing \qquad \text{for every stopping set } S \subseteq R. \tag{6}$$

Thus the problem is to choose a basis of a graphic matroid that is a transversal of the stopping-set family.

Proof. If T misses S, the same stopping set survives in $R \setminus T$, so Lemma 3.4 forbids completion. Conversely, if the restriction to $R \setminus T$ fails to peel, its nonempty canonical residual is a stopping set missed by T. Taking the contrapositive proves the other direction. □

The family in (6) can be exponentially large. The theorem is useful because a failed peel exposes a witness without listing the whole family. Moreover, peeling must not irrevocably reserve a row for lower cancellation. A row peeled during a partial search may still be chosen later as a dual-tree edge. Heredity makes this safe.

3.4 Complete obstruction-guided search

Maintain a selected forest Q of dual edges. Contract it in D and remove its rows from A. Compute the stopping residual K_Q. If it is empty, extend Q to any spanning tree of D and finish. Otherwise choose any nonempty stopping subset $S \subseteq K_Q$, and branch on the edges $e \in S$ that are not loops in D/Q. The child adds e to Q. An inclusion-minimal stopping witness can be sought by successive row removals and re-peeling, but minimality is not required for correctness.

Theorem 3.6 (Complete forest search)**.** *The preceding search is sound and complete for forest-compatible raw completion. Let $s = |V(D)| - 1$ and let w bound the sizes of the stopping witnesses actually used. Its branching depth is at most s, with at most*

$$1 + w + w^2 + \cdots + w^s \tag{7}$$

nodes. If a node, including witness selection, costs $P(N)$ for encoded input size N, the total cost is bounded by (7) times $P(N)$. In particular it is $w^{O(s)}\,\mathrm{poly}(N)$ when $w \geq 2$ and the witness routine is polynomial; for $w = 1$ it is $O((s+1)P(N))$.

Proof. A successful spanning-tree extension of Q must meet any stopping witness S in a row not already in Q. Every such edge is independent of Q, so it appears in one of the stated branches. Conversely each branch retains independence and deletes exactly the rows that its chosen dual edges cancel. Thus branching neither omits a possible completion nor creates a false one.

If $K_Q = \varnothing$, connectivity of D ensures that the forest Q extends to a spanning tree T. The row set $R \setminus T$ is a subset of the already peelable set $R \setminus Q$. Heredity proves completion. The final peel order should be recomputed or checked for this actual T, rather than treating the earlier exploratory order as an unverified certificate.

Each branch increases the forest rank by one. A forest on $s+1$ vertices has at most s edges, proving the depth and node bounds. □

Corollary 3.7 (A local quasi-polynomial regime)**.** *If the source size and coordinate bit lengths are polynomial in a diagram size n, and $s\log(w+1) = O((\log n)^k)$ for fixed k, this restricted forest search has quasi-polynomial bit complexity. The polynomial degree-two-column algorithm in Section 4 requires no small bound on s.*

This is a local certificate-class bound. The source triangulation can have linearly many interior vertices, and no bound on the total hierarchy work follows from it alone.

Proposition 3.8 (Packing obstruction)**.** *Let Q be the currently selected forest, and let $S_1, \dots, S_p$ be pairwise disjoint nonempty stopping sets in $R \setminus Q$. A necessary condition for completion is*

$$\mathrm{rk}_{D/Q}\left(\bigcup_{i=1}^{p} S_i\right) \geq p. \tag{8}$$

For an edge set X, the rank is the number of vertices minus the number of connected components of the graph with edge set X, including isolated vertices and ignoring loops as rank contributions.

Proof. A completion must choose at least one edge from each S_i. Disjointness forces at least p distinct chosen edges in their union. These are part of an independent forest in the contracted graph, so its rank is at least p. □

In particular a stopping singleton is a mandatory tree edge. If mandatory edges form a cycle, no forest-compatible completion exists. A stopping witness whose every edge becomes a loop is the rank-zero case. These certificates use exact graph incidences and exact integer tests.

3.5 Why tree choice matters, even for a ball

Example 3.9 (A geometric two-edge choice)**.** Start with a solid torus, equipped with a fixed meridian belt disk for its one-handle. Choose two disjoint parallel primitive curves on its boundary. Arrange the first to cross the belt disk once. In the annulus complementary to the first curve, a finger isotopy of the second adds a cancelling pair of crossings, so the second crosses three times while remaining embedded and parallel to the first. Attach two two-handles along the curves and cap one of the two resulting sphere boundaries with a three-handle. Before the cap the manifold is $S^2 \times I$; after it the manifold is a ball.

The dual graph has one nonroot vertex and two parallel edges a, b to the root. The raw occurrence matrix is

$$A = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Deleting b leaves a raw singleton and succeeds. Deleting a leaves a stopping singleton and fails. Both choices are valid topological two-/three-handle cancellations. The difference lies entirely in the subsequent restricted raw certificate.

If both primitive curves are given such finger pairs, the matrix becomes $(3, 3)^{\mathsf{T}}$. Each row is a stopping singleton. Their dual edges have rank one, so (8) certifies impossibility in the raw forest class. The manifold is still a ball. Thus even an exhaustive negative answer in this class must remain inconclusive for handlebody recognition.

This is a geometric existence example in a general handle structure. Its actual column totals are four and six, respectively, so it is not a native triangular-face core with the specified cocore. Capping counts preserves the finite cancellation decision, but does not authenticate a native realization. The example establishes that forest choice can matter geometrically; the source tests provide the separate native evidence.

Example 3.10 (Peelable row sets are not a matroid)**.** Consider rows

$$r_0 = (0, 0, 1), \quad r_1 = (0, 1, 1), \quad r_2 = (1, 1, 0), \quad r_3 = (1, 1, 1).$$

The row sets $X = \{r_2, r_3\}$ and $Y = \{r_0, r_1, r_2\}$ both peel and $|X| < |Y|$. But the column totals of $X \cup \{r_0\}$ are $(2, 2, 2)$, and those of $X \cup \{r_1\}$ are $(2, 3, 2)$. Both enlarged sets are stopping. The augmentation axiom therefore fails. One cannot solve the general problem merely by declaring peelable row sets to be a second matroid. The next section obtains a matroid formulation from a genuine structural restriction on column support.

4 A polynomial theorem for sparse active columns

4.1 Replacing peeling by directed reachability

Call a column *inert* if none of its entries is one. Every row restriction still has no unit entry in that column, so it can never supply a unit-column pivot. Its sums on stopping sets can never equal one. It may therefore be ignored for this decision, while its one-handle is retained in the topological count. Assume every non-inert column of A has at most two nonzero rows. Counts can still be arbitrarily large; they are not assumed to be zero/one. Define a directed graph P on the row set R as follows.

- A column supported only on r with $a_{rc} = 1$ makes r an initial seed. A singleton entry of two or more makes no seed.
- For a column supported on $\{u, v\}$, add an arc $v \to u$ if $a_{uc} = 1$, and an arc $u \to v$ if $a_{vc} = 1$.
- Inert columns, including zero columns, make no contribution. Multiple arcs can be merged.

The direction records an implication: deleting the tail makes the head a unit-column row. Let Z be the set of initial seeds.

Lemma 4.1 (Exact reachability model)**.** *For any set Q of rows deleted by top cancellations, the union of Q and all rows subsequently removed by unit-column peeling is exactly the set reachable in P from $Z \cup Q$.*

Proof. Every initial seed can peel. If an arc $v \to u$ comes from a column, then that column has exactly one occurrence on u and no possible other row except v. Once v is deleted, u can peel. Induction on path length therefore shows that every reachable row is removed. The certifying column cannot have been used and removed while its head row was still live: at a unit-column pivot it had no other live owner.

Conversely, a row peeled at any step is certified either by a singleton unit column, making it an initial seed, or by a two-row column whose other row has already been removed. In the latter case the corresponding directed arc gives a reachability step. Induction on the peel sequence proves the reverse inclusion. □

Let the strongly connected components of P be contracted into its acyclic condensation graph. Call a component a *mandatory component* if it is a source of this condensation graph and contains no initial seed. List these components as $S_1, \ldots, S_q$.

Lemma 4.2 (The only mandatory groups)**.** *Deleting Q allows complete peeling if and only if*

$$Q \cap S_i \neq \varnothing \quad (1 \leq i \leq q).$$

The S_i are pairwise disjoint stopping sets.

Proof. A source component with no seed cannot be reached from another component. It therefore needs a row of Q. Conversely every component of a finite acyclic graph is reachable from some

source. If each source either contains a seed or is met by Q, all rows are reachable, and Lemma 4.1 applies.

To see that a mandatory component is stopping, consider a column having an occurrence on a row of S_i. If it has only one row and count one, that row would be a seed. If it has exactly one row in S_i, another row outside S_i, and count one on the inside row, it would give an incoming arc to the source component. Both cases are impossible. If both rows are inside, the total is at least two; if the only inside count is larger than one, the same is true. An inert column can never sum to one. Hence every column has zero or at least two occurrences on S_i. □

4.2 Independent representatives in a graph

Each mandatory component is a set of row labels and therefore a set of edges of D, with ineligible rows represented by root loops. The groups are disjoint. We need one edge from each group, with all chosen edges independent in the graphic matroid. Such a partial forest can be extended to a spanning tree, and any spanning tree meeting the groups contains such a partial forest.

For completeness we state and prove the independent-representatives criterion, a classical result of Rado [10].

Lemma 4.3 (Rado's rank criterion)**.** *Let M be a matroid and $S_1, \dots, S_q$ finite subsets of its ground set. There exist distinct representatives $e_i \in S_i$ forming an independent set if and only if*

$$\operatorname{rk}_M\left(\bigcup_{i\in J} S_i\right) \geq |J| \qquad \text{for every } J \subseteq \{1,\dots,q\}. \tag{9}$$

Proof. Necessity follows by restricting an independent representative set to J. The case $q = 0$ is empty. For sufficiency induct on q. The case $q = 1$ says that S_1 contains a nonloop, which follows from its positive rank.

Suppose a nonempty proper subfamily J is tight in (9). By induction choose independent representatives B for J. They have cardinality and rank $|J|$, so they span $U_J = \bigcup_{i\in J} S_i$. For a set K of remaining groups, contraction by B gives

$$\operatorname{rk}_{M/B}(U_K) = \operatorname{rk}_M(U_K \cup B) - |J| = \operatorname{rk}_M(U_K \cup U_J) - |J| \geq |K|.$$

Apply induction in the contracted matroid to the remaining groups. Their representatives together with B give the required independent set.

If there is no nonempty proper tight subfamily, every such subfamily has rank at least its size plus one. Choose any nonloop $e \in S_q$. For a nonempty set $K \subseteq \{1,\dots,q-1\}$,

$$\operatorname{rk}_{M/e}(U_K) \geq \operatorname{rk}_M(U_K) - 1 \geq |K|.$$

Induction after contracting e supplies the remaining representatives; adjoining e finishes the proof. The contraction treats e as absent from the remaining ground set, so the representatives are distinct. □

Theorem 4.4 (Polynomial forest choice in the two-row regime)**.** *Suppose D is a connected rooted multigraph, and every non-inert column of A has at most two nonzero rows. Compute the mandatory components $S_1, \dots, S_q$ of P. Then the following are equivalent:*

1. *(D, A) has a forest-compatible raw completion;*
2. *the groups S_i have an independent system of representative edges in the graphic matroid of D;*

3. for every set J of mandatory groups,

$$\mathrm{rk}_D\left(\bigcup_{i\in J} S_i\right) \geq |J|. \tag{10}$$

These conditions can be decided in polynomial time. A positive certificate is a spanning tree with an explicit complete peel order. A negative certificate is a subfamily J violating (10), together with the mandatory-component data, which can be recomputed independently.

Proof. Lemma 4.2 reduces completion to choosing a spanning tree meeting each mandatory group. Such a tree contains independent representatives. Conversely an independent representative set extends to a spanning tree, which meets every group and therefore succeeds. Lemma 4.3 proves the equivalence with the rank inequalities.

For the algorithmic assertion, restrict the ground set to the union of the groups. Intersect the graphic matroid with the partition matroid allowing at most one edge from each group. A common independent set of size q gives the representatives. Matroid intersection has a polynomial augmenting-path algorithm with independence oracles [2]. Here the oracles are elementary: a disjoint-set forest detects graphic cycles, and color counts test the partition condition. The residual exchange graph of a failed maximum intersection produces a rank defect; equivalently the matroid-intersection min–max theorem and Lemma 4.3 supply the subfamily J. The delivered implementation extracts such a witness and its checker verifies the inequality directly, without trusting an optimality assertion. □

4.3 Explicit algorithm and complexity scope

The algorithm first scans columns to detect the unsupported case of an active column with more than two incident rows. It then computes seeds, arcs, strongly connected components, and mandatory groups. These steps take linear time in the matrix incidence size, with standard adjacency-list graph algorithms.

Starting from the empty common independent set, the matroid-intersection algorithm builds an exchange graph. Its vertices are candidate edges. Arcs encode exchanges that preserve one of the two matroids, and endpoints encode single-element additions that preserve the other. A shortest augmenting path toggles membership and increases the common independent set by one. If no path exists, the reachable set yields the min–max certificate. There are at most $q \leq |R|$ augmentations. Even a conservative implementation using $O(|R|^2)$ oracle calls per augmentation and rebuilding a graphic oracle in $O(|R| + |V(D)|)$ operations is polynomial:

$$O(I + |R|^3(|R| + |V(D)|)),$$

where I is the occurrence-input size. No enumeration of all subsets J in (10) is required. The implementation and measured operation counts are documented in Sections 5 and 6.

The column restriction is a geometric crossing restriction in an authenticated core, not a bound on algebraic rank or on the number of vertices of D. Large counts on either of the two rows are permitted. For example, an abstract solver input column $(1, 2^{1000})^{\mathsf{T}}$ contributes just the arc from the second row to the first. A column with entries $(2, 2, 2)$ is inert and harmless for the restriction. A column with entries $(1, 1, 1)$ is outside the theorem. These large-count examples concern the generalized finite solver, not the original native columns bounded below.

4.4 Why shortest augmentation works, and where the obstruction comes from

Here is a self-contained justification for the specialized exchange step. Restrict to the union G of the mandatory groups and maintain a rainbow forest F. The exchange graph has sources $e \notin F$ for which $F + e$ is a forest and targets $e \notin F$ of an unrepresented group. For $f \in F$ and $e \notin F$ it has arcs

$$f \longrightarrow e \quad \text{if } F - f + e \text{ is a forest}, \qquad e \longrightarrow f \quad \text{if } F - f + e \text{ is rainbow}.$$

Take a shortest source-to-target path $e_0, f_1, e_1, \ldots, f_\ell, e_\ell$. None of its intermediate outside vertices is a target, so e_{i-1} has the group of f_i, whereas e_ℓ has a new group. The symmetric difference is therefore rainbow and has $|F| + 1$ edges.

For graphic independence, represent edges by vertex-incidence vectors over $\mathbb{F}_2$. The forest vectors are independent. The source e_0 is outside their span. Every later e_i is not a source, since otherwise its suffix would shorten the path. It therefore has a unique fundamental cycle in F. Its expansion in the forest basis has coefficient one at f_i and coefficient zero at each f_j with $j < i$; a nonzero earlier coefficient would supply the shortcut $f_j \to e_i$. Modulo $F \setminus \{f_1, \ldots, f_\ell\}$ these expansions form a triangular matrix with unit diagonal. They are independent, and adding e_0, outside the original span, preserves independence. The augmented set is a larger rainbow forest.

If no target is reachable, let W be the reachable set of edge labels. Every group appearing in W is represented by an edge of $F \cap W$: an unrepresented reachable group would give a target, and an outside edge in a represented group has an arc to that group's selected edge. Thus exactly $|F \cap W|$ groups appear in W. Each outside edge not in W is spanned by $F \setminus W$: it is not a source, and its fundamental cycle has no reachable selected edge, which would give an incoming exchange arc. Hence

$$\mathrm{rk}_D(G \setminus W) = |F \setminus W|.$$

Let J consist of the groups with no edge in W. If $|F| < q$, then

$$\mathrm{rk}_D\left(\bigcup_{j \in J} S_j\right) \le |F \setminus W| = |F| - |F \cap W| < q - |F \cap W| = |J|.$$

This supplies the claimed negative certificate directly. The checker only needs the stopping sets and the rank inequality, not the internal exchange graph.

4.5 A geometric bound isolating the remaining native case

Proposition 4.5 (Three occurrences per native cocore)**.** *For the unsigned matrix of original native core two-handles against native one-handle cocores,*

$$\sum_r a_{rc} \le 3 \qquad \textit{for every column } c.$$

The same holds after contraction of a fixed lower zero-/one-handle forest and any dual-tree row deletion. It is not asserted after adding boundary relations or performing later handle slides.

Proof. Work in one abstract triangular source face before quotient identifications. Its disjoint normal arcs have endpoints on distinct local sides. Every complementary disk region meets each of the three sides in at most one gap. This follows inductively: an additional arc splits one region, divides its intervals on the two endpoint sides between the children, and leaves each other side interval wholly in at most one child. Initially the assertion holds for the uncut triangle.

The cocore of a native one-handle is one such face region. Each geometric traversal by a core two-handle corresponds to an actual boundary-gap occurrence on that cocore, of which there are at most three. Different occurrences remain counted even if quotient identifications give them the same global edge or handle label. Removing parallelity two-handles can only decrease the count. Lower forest contraction deletes its columns and introduces no crossings of retained cocores. Dual-tree normalization deletes rows while retaining their attaching charts. □

Consequently, up to permutation, every nonzero native column has positive-entry multiset

$$(1),\ (2),\ (3),\ (1,1),\ (1,2),\ \text{or } (1,1,1).$$

The last shape is the only one outside Theorem 4.4. For its three rows u, v, w, peeling gives the two-premise implications $\{u, v\} \to w$, $\{u, w\} \to v$, and $\{v, w\} \to u$. This is a precise point at which ordinary directed reachability ceases to suffice.

Proposition 4.6 (Parity compatibility with the dual graph)**.** *Use the actual unsaturated source counts in this statement. Before top cancellations, the dual edges whose entries in a fixed source column are odd form an even-degree subgraph of D. The same statement holds for the retained rows in D/Q after contracting a selected dual forest Q.*

Proof. Work over $\mathbb{F}_2$. In the handle chain complex, the boundary of a three-handle records the actual cap incidences of its two-handles modulo two. Thus ∂_3 is the incidence matrix of D with the formal root row omitted, transposed to the handle-boundary convention. The corresponding row of ∂_2 is the unsigned source column modulo two. The identity $\partial_2\partial_3 = 0$ implies even degree at each three-handle vertex. Loops contribute twice. The root also has even degree, since the sum of all graph degrees is even. Contraction of a nonloop combines its endpoint parity equations and deletes that edge; an even subgraph remains even in the contracted multigraph. Iterate along Q. □

A native $(1, 1, 1)$ column is therefore a three-edge mod-two cycle: a triangle, a parallel pair together with a loop, or three loops. A singleton odd entry must label a dual loop. These are necessary geometric compatibility conditions, not a realization theorem for arbitrary matrices. The nonmatroid example in Section 3 already has column totals $(2, 3, 3)$. It can also satisfy the parity condition: make r_0, r_2, r_3 three parallel root-to-vertex edges and r_1 a loop. Its three columns are then even subgraphs. Thus neither of these two necessary conditions restores the matroid exchange axiom. No native triangulated realization of that example is claimed.

4.6 The boundary between polynomial and general search

Theorem 4.4 does not establish an intrinsic complexity dichotomy for all handle structures. We do not prove the unrestricted problem NP-hard, and an improved algorithm may exist for other classes. The exact obstruction-guided procedure remains available when the column restriction fails. Its canonical residual and graphic contraction can also expose a smaller two-row instance later in the search.

A practical implementation can therefore use the polynomial solver first, then use the general search with an explicit budget. Positive output always carries a tree and checked peel order. Exhaustion under a budget must produce an inconclusive status; it must not be confused with a negative rank certificate or an exhaustively checked search tree.

5 Implementation, certificates, and integration

5.1 A small extension to the native entry point

The production-facing change is deliberately confined to a supplied-cut operation. Four new modules implement geometric graph compilation, its independent replay, the completed boundary-kernel branch, and its replay. Two existing entry points acquire a dispatch to the new branch. The graph and reduction routines receive the authenticated geometry that the caller has already constructed. Consequently their introduction does not repeat the weighted-orbit discovery that dominates some cuts.

The producer follows this sequence:

1. Compile the actual cap endpoints of every source core two-handle. Maintain a dense array of source vertex flags; exposed flags receive the root label. No array is indexed by a normal coordinate value.

2. Find a rooted breadth-first tree. Delete its two-handles and all three-handles, preserving the surviving two-handle order and assigning consecutive reduced identifiers. Keep the lower charts, ports, and source coordinates intact.

3. Construct the existing ordered parallelity boundary skeleton, using the remainder of the caller's orbit budget. Run the existing finite boundary kernel on the reduced geometry and that skeleton.

4. If the kernel completes, return its exact component census and, when requested, the composite certificate. Otherwise return an inconclusive result with the recorded stage and reason.

The shipped native producer uses a deterministic breadth-first tree. The more sophisticated forest solvers are delivered as separate research modules and are used in the matrix audit. They are not silently inserted into the production entry point. On the ten affected corpus cuts, exhaustive fixed-kernel analysis finds no additional positive census beyond the seven obtained by this simpler native branch.

5.2 The composite certificate and trust boundary

The new outer schema is `normal-cut-handlebodies-v6`. Its exact fields are `schema`, `parallelity`, `boundary`, `dual`, `kernel`, and `census`. The two source packets are the existing normal-parallelity and ordered-boundary certificates. The new inner schema is `source-core-dual-forest-v1`, with just `schema` and `forest`; the latter is a list of original two-handle identifiers.

The independent outer checker first authenticates both source packets from the supplied triangulation and normal vector. It then reconstructs cap endpoints and checks the supplied tree by union–find. It requires exactly s distinct, valid, acyclic edges and verifies connection to the root. It constructs the retained chart separately, replays the existing boundary kernel, and compares the claimed census with the reconstructed one. The independent dual checker imports no graph compiler or forest producer. It uses its own endpoint table, a different root label, and independent connectivity bookkeeping. Sharing the mathematical endpoint convention is necessary; sharing a producer's unchecked graph would not be sufficient.

The inner checker is intentionally a checker for *authenticated* geometry, not a substitute for the source validator. Calling it directly on an arbitrary dictionary does not authenticate a three-manifold. This precondition is documented in the modules and is enforced by the outer entry point. The

contract excludes false inferences from an algebraic incidence matrix: a coefficient of ± 1 is not accepted as a geometric two-/three-handle cancellation without the actual cap occurrences.

Native source integers use the repository's exact integer codec. Large binary multiplicities survive replay and census comparison without a machine-word truncation. The tree itself uses only $O(t)$ source labels. Normal weights occur only in endpoint arithmetic and in the already existing source and bundle packets. The statement that the new stage is polynomial in encoded size is not a statement that every previous stage's implementation has the same tight bound.

5.3 Failure and resource semantics

Successful native output is `COMPLETE`; all uncompleted native paths are `INCONCLUSIVE`. In particular, inability to root a supplied dual graph is conservative even though Theorem 2.6 rules it out under the native contract. A stalled boundary kernel is also conservative. Neither status asserts that the source knot is nontrivial.

The callback used by the existing pipeline is propagated through graph construction, adjacency traversal, boundary processing, and replay. The outer verifier's `max_operations` parameter bounds the existing orbit-proof operation records; it is not a universal bound on every Python instruction or on arbitrary malformed input length. The callback remains the surrounding cancellation mechanism. A hard total verification budget would need a separate, consistently charged policy across all imported checkers. The tests preserve the current semantics rather than inventing an incompatible interpretation for this one branch.

The finite graph/matrix solvers use `SUCCESS`, `NO_FOREST`, and `INCONCLUSIVE`. Their negative statement has a different and strictly specified meaning: no compatible tree completes raw cancellation in that supplied model. They can certify this by a disconnected graph cut, a disjoint stopping-set rank defect, or a complete obstruction search tree. A disconnected finite input is useful for checker testing; it cannot arise from the admitted native source graph. Work or node exhaustion, and the general Python producer's interpreter recursion limit, give only `INCONCLUSIVE`. The complete mathematical search theorem refers to the finite procedure with sufficient resources.

5.4 The separate finite graph/matrix implementation

The source object has an integer vertex count, an integer column namespace, one sparse occurrence row per two-handle, and one graph edge or `null` per row. Vertex zero is the root. A `null` row is not eligible for top cancellation; this covers the extra annular relations in the boundary kernel. Equivalently, the theory may give every such row an artificial root loop. This convention allows the same stopping criterion to include both deletable source relations and nondeletable interface relations.

Counts are replaced by their values capped at two. Sparse dictionaries retain only named columns, so a very large binary column namespace does not allocate that many entries. Input validation rejects duplicate incidences and malformed labels. The graph checker also detects an omitted isolated vertex without allocating a declared, potentially enormous namespace. These details keep validation tied to the supplied encoding.

The polynomial producer computes initial unit seeds and directed arcs, then the source strongly connected components. A shortest-path exchange algorithm constructs a rainbow graphic forest. The general producer computes the stopping core, optionally shrinks a witness to inclusion minimality, contracts a chosen graph edge, and branches only on eligible nonloops. Singleton choices are forced. Every positive result includes the final spanning tree and a peel order on the immutable source

rows. The verifier checks each advertised unit-column pivot against those rows, without running the producer's peeling queue or exchange search.

For a rank obstruction the checker verifies that the listed row groups are disjoint, nonempty stopping sets; computes the rank of their eligible edge union; and checks that this rank is smaller than the number of groups. It does not need to trust that the producer identified strongly connected components correctly. For a negative search tree it checks every node's stopping set and exact coverage of the eligible branches, including the accumulated forest contractions. This distinction between discovering and checking a witness is central to the implementation.

5.5 Verification strategy

The native focused tests cover the source orientation convention, ordinary and loop edges, self-identifications, multiple interior vertices, geometric port retention, exact Euler characteristic, source and certificate mutations, older-schema dispatch, resource callbacks, one-sided normal surfaces, and very large binary multiplicities. An independent cap oracle walks source local incidences on all 106 corpus cuts. Producer-disabled tests ensure that replay does not call the new discovery routines.

The finite focused tests include true augmenting paths of length three, rank defects requiring a proper subfamily of source components, exhaustive negative proof-tree coverage, loops and parallel edges, ineligible rows, nonunit counts, huge sparse namespaces, and adversarial certificate mutations. The larger oracle audit separately enumerates spanning trees and nonempty stopping subsets on small cases. This tests both positive and negative claims against a structurally different decision procedure.

These are executable regression and cross-checking tests, not a formal proof assistant development. The manuscript supplies the mathematical proofs, and the checkers reduce the executable trust surface. Regina [12] supplies a further, independently implemented topological comparison for the new native positive censuses.

5.6 Integration boundaries

The package targets the exact source revision `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`. Apply its patch to that revision, run the focused tests and source replay, and then rebase through the current entry points if the repository has advanced. The small dispatch changes should be reviewed especially carefully against newer certificate schemas or budget semantics. The patch does not revise the full recognizer's verdict logic, widen the native source validator, or assert completeness of the boundary kernel.

The standalone forest module is ready for experimentation but remains a separate integration decision. A future production importer must bind its matrix to authenticated geometric words and label every nondeletable relation correctly. The matrix audit included here demonstrates that binding for the current fixed boundary kernel; it does not justify reusing a cached matrix after an unrecorded change of lower forest, interface attachment, or handle slide.

6 Experiments and what they establish

6.1 Pinned corpus comparison

The baseline is the supplied-cut entry point at `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`, loaded from that exact Git object. Its dependencies are held at the same revision. The corpus is the 106-case `marked_parallelity_research/corpus.json`. Its SHA-256 digest and the baseline

entry-point digest are recorded in `results/native/native_audit.json`. Every result records a case label, statuses, timing scope, census when present, and certificate information.

Table 1: Native supplied-cut census coverage at the pinned revision.

Outcome	Baseline	With dual normalization
Complete censuses	85	92
Inconclusive cuts	21	14
Earlier complete censuses preserved	–	85
Successful independent certificate replays	–	92
New censuses agreeing with Regina	–	7

The coverage rises from 80.19% to 86.79%. Seven of the 21 previously inconclusive cuts become complete, reducing that count by one third. This is a measured coverage improvement on supplied cuts. It is not a measured speed-up of the end-to-end knot recognizer, and a previous inconclusive return is not a valid fast baseline for a newly completed query. The README's older success count is stale; the comparison uses the measured 85 rather than that prose count.

Table 2: All seven newly certified cuts. The notation g^m means m components of genus g, not exponentiation.

Corpus label	Genus census
`pilot/interior_1_2/0`	1^2
`pilot/interior_2_3/0`	1^2
`pilot/interior_3_5/0`	1^1
`pilot/finite_trefoil_interior/0`	2^1
`interior_1_2/mixed`	0^2
`interior_3_5/mixed`	0^2
`finite_figureEight_interior/mixed`	$0^1, 2^2$

The package retains the seven full source/certificate packets. The audit's sorted-key JSON encodings of the seven composite certificates alone range from 11,631 to 1,069,420 bytes. The last, larger certificate mainly reflects existing source and boundary proofs, not a large dual tree. All seven affected native trees have one edge in this corpus. Separate subdivision fixtures exercise many three-handles, so the tests of the graph stage are not confined to one-edge cancellation.

6.2 Which matrices need more than the polynomial theorem?

The matrix audit first fixes the native lower zero-/one-handle spanning forest and reads the original source two-handle cycles. Of the 106 matrices, 12 have maximum column support zero, seven have maximum support one, 62 have maximum support two, and 25 have maximum support three. Thus 81 of 106 satisfy the two-row condition before adding interface relations. The audit also checks that the total raw occurrence count of every original source column is at most three.

A separate source-parity audit uses uncapped attaching-cycle counts and an independently oriented cap oracle. It checks 3,239 original one-handle columns and 783 source two-handle rows across the 106 sources, including columns later removed by the lower forest. It finds no violation of either Proposition 4.5 or Proposition 4.6. Of 74 triple-unit columns, seven give a parallel pair plus a loop and 67 give three loops. The corpus has at most one interior vertex per source, so it cannot test genuine dual triangles. Three separately labelled supplementary stellar sources with two, four,

and eight interior vertices provide respectively three, nine, and 25 triangle columns; all pass the same independent check. These supplemental sources are not counted as additional members of the 106-case coverage corpus.

There are ten cuts with three-handles. Seven of their bare core matrices are in the polynomial regime, and the polynomial solver finds and verifies a successful tree in all seven. The other three bare core matrices also have successful trees, found and independently checked by the general stopping search. *All ten bare cores are therefore solvable in the fixed raw model.* This does not settle the actual cuts, because the parallelity attachments still contribute relations.

The audit next inserts the same finite annular relations used by the boundary kernel. Four of these ten augmented matrices satisfy the polynomial hypothesis. General forest search succeeds on exactly the seven cuts of Table 2; it certifies `NO_FOREST` on the other three. In each negative case there is a stopping witness consisting entirely of nondeletable interface rows: Each displayed set is

Table 3: Obstructions that no change of dual tree can remove in the fixed augmented matrix. Row identifiers refer to the saved source problem.

Corpus label	Stopping rows	Eligible rank
`pilot/finite_figureEight_interior/0`	$26, \ldots, 32$	0
`interior_2_3/mixed`	$11, \ldots, 14$	0
`finite_trefoil_interior/mixed`	16	0

a single nonempty stopping set with no eligible dual edge. It is therefore already an obstruction by Theorem 3.5; no exponential negative search is needed to check it. This evidence changes the practical research priority: trying additional dual trees cannot resolve these three fixed kernels. The attachment treatment or the permitted lower-handle moves must change.

The fourteen remaining native inconclusive cases divide into eight raw cancellation stalls, three stalls in the boundary kernel after dual reduction, and three raw ball-cancellation obstructions. The exact reason codes are retained in the JSON audit. These are pipeline outcomes, not a classification of the underlying knots.

6.3 Complete-query timings on a parallel-surface family

We compare two routes to the same component genus census for multiples of a one-sided normal surface in a finite solid-torus fixture with an interior vertex. The native timed route includes certificate production and independent replay. The Regina route includes source conversion, cutting along the normal surface, constructing connected components, and exact handlebody recognition of those components. It materializes the cut triangulation; the native route retains multiplicities in its compressed census. JSON serialization, interpreter startup, and import time are outside both timed arms.

There are five randomized paired rounds at each size, seed 20261012, using `perf_counter`. The recorded environment is Python 3.12.14, Regina 7.4, Linux 6.18.44 on x86-64. Table 4 reports median total times and the median of the five within-pair time ratios. The latter need not equal the ratio of the two marginal medians. Raw rounds and stage timings are included; five rounds are too few for strong distributional claims.

Regina is much faster on small inputs, by almost a factor of nineteen at four copies. The routes are close at 256 copies, and the native route is about 4.8 times faster at 1,024 copies. The nonmonotone native timings show fixed costs and runtime variation; they are not a fitted asymptotic law. The asymptotic statement for the new graph stage follows from Theorem 2.6,

Table 4: Complete supplied-cut queries, milliseconds. A ratio above one favors the native certified route.

Copies	Components	Cut tetrahedra	Native	Regina	Paired ratio
1	1	91	4.056	0.895	0.225
4	3	151	26.507	1.420	0.0528
16	9	391	27.514	3.014	0.107
64	33	1,351	28.289	8.825	0.317
256	129	5,191	40.501	42.282	1.044
1,024	513	20,551	35.723	169.545	4.824

not from extrapolating this table. The comparison also does not rule out a different compressed implementation using Regina components or alternative exact recognition procedures.

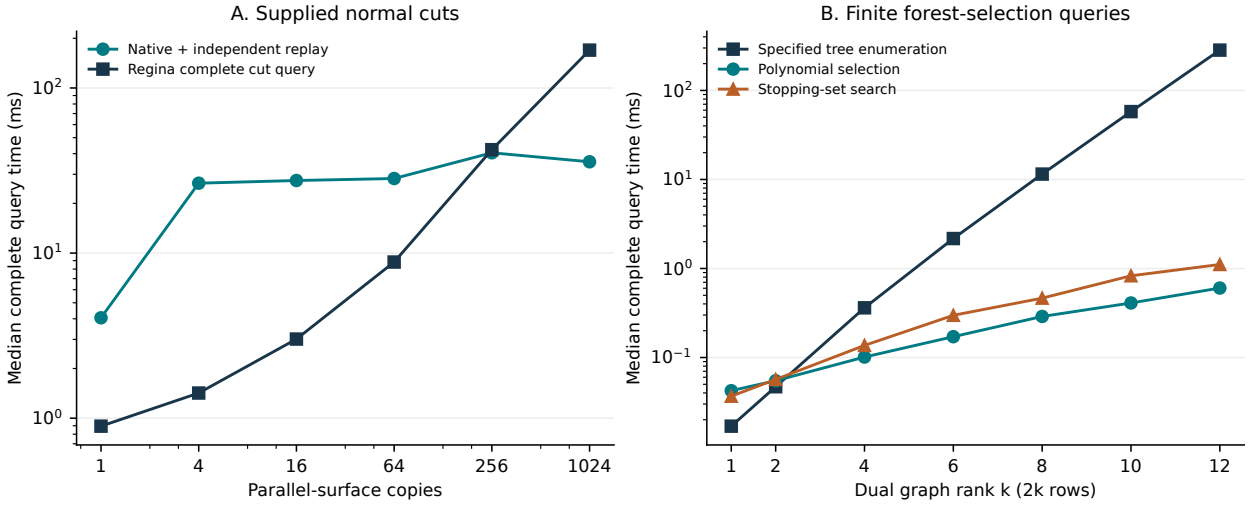


Figure 1: Recorded complete-query medians. Left: supplied native cuts, including independent replay, against the stated Regina route. Right: the finite forest-selection family, including the same positive checker in every arm. The two panels have different inputs and must not be combined into a knot-recognition speed-up.

6.4 Finite decision audit and a controlled enumeration comparison

The independent finite audit checks 33,995 graph/matrix problems. Of these, 26,995 are exhaustive enumerations in the stated small families and 7,000 are seeded random cases. The separate oracle finds 4,554 feasible and 29,441 infeasible instances. Every general-search certificate replays. The polynomial solver completes and verifies 23,849 instances; it declines the other 10,146 because an active column has more than two owners. These are finite graph/matrix tests, not 33,995 normal-surface examples.

For a controlled comparison, take a chain of k dual links, each replaced by two parallel edges a_i, b_i. Column i has its sole occurrences on these two rows, with counts 1 and 2, equivalently the capped form of counts 1 and 3. There are 2^k spanning trees. Every $\{b_i\}$ is a mandatory stopping singleton, so the sole successful tree is $\{b_1, \ldots, b_k\}$. The specified include-before-exclude enumerator, with a_i before b_i, reaches this tree last and tests exactly 2^k complete trees. Stopping search makes k forced choices; the polynomial solver finds k source-component representatives. This proves

an exponential separation from that explicit enumeration policy, not a lower bound against all algorithms.

Table 5: Finite forest-selection complete queries, median milliseconds over five paired rounds. Every positive arm includes independent replay.

k	Trees tested by enumeration	Enumeration	Polynomial	Stopping search
1	2	0.0169	0.0424	0.0368
2	4	0.0470	0.0549	0.0567
4	16	0.3616	0.1012	0.1367
6	64	2.1693	0.1713	0.2978
8	256	11.4927	0.2890	0.4642
10	1,024	57.8562	0.4096	0.8278
12	4,096	283.8067	0.6030	1.1089

At $k = 12$ the polynomial query is about 470.6 times faster than the specified enumerator. At $k = 1$ it is about 2.5 times slower. An identical enumerator control gives an A/A median-time ratio of 0.987 at $k = 12$. The larger polynomial-only run reaches $k = 128$ in 224.3 milliseconds, including replay. The hypothetical 2^{128} enumeration count there is a formula, not a measured enumeration run. Structural visit counters are included for reproducibility; their totals are not CPU instruction counts and should not be equated across implementations.

6.5 Stress tests and remaining measurement limits

The native focused suite constructs successive interior subdivisions, exercising up to sixteen interior vertices, as well as normal multiplicity 2^{4096}. It checks a compressed exact census and replay without enumerating that many normal pieces. The large-integer case tests the encoded interface; it does not by itself prove a uniform time bound for all large-coordinate inputs. The general finite suite separately tests huge sparse namespaces and resource exhaustion.

The full repository regression suite passes 1,688 tests in the recorded run (374.807 seconds). The standalone finite suite adds 18 focused test methods; the ten new native methods are included in the 1,688. The first sparse checkout lacked historical report fixtures needed by older tests; those dependencies were restored before the final regression run. This setup failure is recorded separately from the final run and was not counted as a source-code defect or hidden by changing the tests. No peak-memory comparison, realistic knot-distribution benchmark, or end-to-end diagram-recognition timing is claimed in this work.

7 Toward a quasi-polynomial hierarchy

7.1 What this work removes from the complexity budget

There are three different sizes that must not be conflated: the original diagram size n, the encoded size of a current triangulated or uniform handle state, and the expanded weight of a normal surface in that state. The new normalization controls an operation in the second size while avoiding dependence on the third as a unary parameter. It does not yet control how many states are discovered from the first.

In particular, a source with many interior vertices no longer forces an immediate failure of the finite census method. Theorem 2.6 removes all its top handles in polynomial encoded time, with no assumption that their number is logarithmic. The polynomial selection theorem also removes

a possible enumeration of dual trees in its stated column regime. For a general matrix, however, the complete selection search retains its w^s parameter dependence. The native geometric bound reduces its original-column difficulty to cycle-compatible triple-unit columns, but the augmented interface matrix can introduce additional difficulty.

The global composition issue can be stated without hiding an oracle. Suppose a hierarchy discovery procedure makes at most $Q(n)$ state visits, including revisits and rejected branches, every state and emitted certificate has encoded size at most $S(n)$, and the total discovery, transition-generation, local processing, and replay work charged to each state visit costs at most $S(n)^d$, for a fixed constant d. A sequential implementation then costs at most $Q(n)S(n)^{O(1)}$, plus the explicitly accounted input and output work. If $Q(n) = 2^{O((\log n)^3)}$ and $S(n) = n^{O(1)}$, this is the announced quasi-polynomial scale. This elementary composition observation is a budgeting rule, not a new recognition theorem.

Lackenby's bound $L(g+1)^L$ gives the same scale if a suitable implemented hierarchy can establish $L = O((\log n)^2)$, $g = n^{O(1)}$, and polynomial encoded work at every node. The linear hierarchy bound in the 2026 preprint, by itself, does not imply that logarithmic-square value of L for arbitrary input. Likewise a theorem that a small certificate exists does not bound the work of finding it. These are the principal gaps to be closed, rather than details that can be suppressed in a local runtime.

Table 6: Complexity obligations and their present status.

Obligation	Status in this work
Compressed cutting	Established in prior literature; the native pipeline implements a source-certified form.
Remove native core three-handles	Proved and implemented in polynomial encoded time for the admitted source contract.
Choose a tree for raw completion	Polynomial in the sparse active-column case; complete parameterized search in general.
Transport recorded bundle ports	Proved and implemented for the current exposed zero-/one-handle port charts.
Transport arbitrary hierarchy patterns	Open implementation and proof obligation when patterns cross the cancellation support.
Find suitable surfaces efficiently	Not solved by this operator; uniform discovery costs must be bounded.
Control state size and hierarchy depth	Must be established relative to the original diagram, including all resets and certificates.
Decide every terminal state	The current raw kernel is sufficient but incomplete; fourteen corpus cuts remain inconclusive.

7.2 Priority 1: remove the certified interface obstructions

Question 1: which additional geometric moves discharge the three saved annular stopping sets? The negative packets make this question sharply testable. On the three cuts in Table 3, altering only the dual spanning tree cannot work. A next implementation should keep the authenticated source and change a precisely named operation: the presentation of a bundle attachment, a permitted disk-bundle absorption, or a certified lower-handle move. It should then show how the formerly nondeletable stopping rows become removable or disappear under an actual geometric equivalence. The success criterion is a new replayable genus census, not merely an algebraic simplification of the row matrix.

Question 2: can an interface obstruction be localized to a bounded subsurface? The saved stopping sets have sizes seven, four, and one. Trace their annular curves through the ordered boundary skeleton and determine the smallest marked subsurface carrying them, together with its complementary attachment data. A useful theorem would show that a certified local replacement has cost polynomial in the subsurface's encoded complexity and preserves its gluing map relative to the complement. A graph support or homology calculation alone cannot guarantee this embedding statement.

Question 3: can primitive but nonunit attaching curves be recognized with a compressed geometric certificate? The ball example shows why unsigned raw counts can stall on a primitive curve. One candidate extension is a sequence of boundary isotopies or slides whose bands are specified by compressed paths and whose effect on every other ordered attaching curve is independently replayed. The target is not unrestricted free reduction of a word: cancelling algebraic inverse letters can conceal geometric intersections with other bands. A successful theorem must bound both the move sequence and the description of the transported curves. The singleton annular obstruction is a useful first native test case.

7.3 Priority 2: solve the remaining native column geometry

Question 4: is forest-compatible peeling polynomial for cycle-compatible triple-unit columns? Propositions 4.5 and 4.6 show that the original native columns lie in a much narrower class than general sparse matrices. The remaining active shape is $(1, 1, 1)$, and its edges form a triangle, a parallel pair plus a loop, or three loops. The question is whether these geometric restrictions allow an exact extension of the source-SCC method. They do not restore matroid exchange, as the explicit counterexample shows. An affirmative algorithm therefore needs additional structure; a negative complexity result would require a reduction that respects these constraints, preferably with a genuine native realization.

Question 5: is the number of triple-unit columns a useful parameter? Let k count the original columns of shape $(1, 1, 1)$ after the chosen lower forest. A strong target would be an algorithm of cost $2^{\mathrm{poly}(k)}\,\mathrm{poly}(m, I)$, or an explicit smaller parameter based on the interaction graph of those columns. This would complement the current $m^{O(s)}$ search when s is large but k is small. The immediate combinatorial model consists of ordinary reachability plus two-premise activation rules. Treating a two-premise rule as either of its individual arcs is unsound; both prerequisites must be discharged, and the initial deletion set must remain a graphic forest.

Question 6: can graphic rank be used earlier inside the activation process? For ordinary source components, disjointness converts all obligations into a partition matroid. Ternary activation may create overlapping obligations. One approach is to maintain certified disjoint stopping families and their contracted ranks throughout the search, extracting short rank defects before branching. Another is to identify separators in the joint dual/dependency incidence graph and combine states recording both connectivity and activation. The research obligation is an explicit state-count bound; declaring a separator small without bounding all boundary partitions does not establish quasi-polynomial complexity.

Question 7: what exactly does changing the lower forest buy? The complete search theorem fixes the zero-/one-handle spanning forest. Changing it changes which cocores survive,

and can change the available raw pivots. Study paired choices of a lower forest and an upper dual forest while retaining the source geometric cycles. A first experiment should exhaust both choices on the smallest authenticated failures, record whether success depends on the lower choice, and save a replayable certificate. A theorem that bounds a representative family of lower forests would be more valuable than unrestricted double enumeration.

7.4 Priority 3: make hierarchy interfaces compositional

Question 8: can arbitrary marked boundary patterns be transported through the cancellation balls with bounded encoding growth? The present proof fixes exposed native ports. In a general sutured or patterned hierarchy, marked curves may cross the support of the selected top cancellation. The next geometric compiler must specify how such curves move across the boundary disk E and the replacement collar, prove that the cyclic order and essential intersections are preserved, and bound the growth of the compressed description. A useful target is a polynomial bound in the input pattern length and source chart size per complete forest reduction, rather than a separate expansion at every three-handle.

Question 9: can support-stratum reuse extend beyond the dual tree without losing source authentication? Corollary 2.7 identifies a valid reuse opportunity: the dual graph and selected tree depend only on zero versus positive source edge weights. The full boundary words and bundle multiplicities need not be constant on that stratum. Seek a reusable symbolic template for the retained chart whose variable lengths and endpoint offsets are explicit arithmetic expressions in normal coordinates. Each evaluation must still authenticate the relevant source equalities. This would remove repeated structural work while keeping multiplicity-dependent attachments honest.

Question 10: can local normalization be made confluent at a useful level of abstraction? Different dual trees produce homeomorphic cores with potentially different raw cancellation success. Full geometric confluence is therefore too strong for the present representation. A more useful target is a canonical summary of the remaining obstruction class, perhaps in terms of marked boundary components and a certified family of stopping/rank witnesses. The summary must preserve the information needed by subsequent cuts; equal Euler characteristic, equal genus totals, or equal homology alone does not suffice.

7.5 Priority 4: close the discovery and global-size gaps

Question 11: which supplied surfaces can be found with a proved encoded-time bound? Start with a restricted source class where the existence theorem supplies a normal disk or hierarchy surface of bounded combinatorial type. Specify the actual discovery algorithm, the bit lengths of its intermediate coordinates, and the total unsuccessful work before a certificate is found. A normal-coordinate vector supplied by an experiment is an input to this work's operator, not a discovery algorithm. A promising bridge to the hierarchy programme is a certified search over compressed surface templates whose count can be bounded independently of expanded weight.

Question 12: can one charge every reset to a monotone global quantity? The incoming reports already discuss causal reset budgets and cographic persistence. The next step should connect those budgets to the actual geometric importer and this top normalization. If an attachment change invalidates a previous certificate or reintroduces branching, the reset must have a reason recorded in a global potential. Seek a statement bounding the total number and cumulative encoded cost of

resets along every search path. Independent local size bounds do not prevent an algorithm from returning to exponentially many distinct states.

Question 13: can the total certificate, including all rejected branches, be kept quasi-polynomial? Positive compressed certificates are often small while a search transcript is large. Distinguish a certificate needed to verify the eventual answer from evidence needed to justify exhaustive failure of a branch. The rank defects here illustrate a beneficial compression of negative search. Identify analogous short obstructions for other hierarchy operations, and prove that their discovery cost is also controlled. A concise negative certificate discovered only after exponential search does not resolve the runtime question.

Question 14: which terminal procedure should replace raw cancellation when it is inadequate? The present kernel gives useful sufficient certificates, but the geometric ball example proves its incompleteness in the broader handle category. An eventual complete hierarchy needs a terminal method with a proved worst-case budget for every permitted terminal state. Candidate directions include restricted geometric simplification, a certified comparison to standard handlebodies, or a different hierarchy refinement. The choice should be evaluated against the actual terminal-state contract rather than the full unrestricted handlebody problem.

7.6 Priority 5: verification and credible performance measurement

Question 15: what is the smallest independent trusted kernel? The current replay reduces dependence on discovery code but remains Python and relies on the existing source geometry checker. A serious next step is a written specification for integer decoding, source incidence, tree reduction, and ordered boundary transport, followed by a small reference checker or proof-assistant formalization. The parity constraint is a useful independently derivable invariant, but should be checked on uncapped source counts: capping three to two changes parity.

Question 16: where is the practical crossover on nonconstructed inputs? The reported family isolates compression and the finite enumeration family isolates tree choice. Neither is a realistic input distribution for all unknot recognition. Extend measurement to a stratified native corpus, reporting triangulation size, coordinate bit length, number of interior vertices, number of triple-unit columns, interface size, genus census, certificate size, wall time, and peak memory. Compare complete queries with randomized paired order and report regressions. For an end-to-end claim, include preprocessing, surface discovery, all rejected cuts, final certificate production, and replay.

7.7 Suggested next sequence of work

The most informative immediate sequence is to resolve the three saved interface obstructions with a named geometric extension; characterize cycle-compatible ternary activation on native cores; and implement marked pattern transport for the resulting reductions. In parallel, one should build an accounting layer that records discovery costs and state sizes against the original diagram. These tasks turn the current local theorem and measured coverage improvement into reusable components of a hierarchy argument. The general quasi-polynomial recognition theorem remains the goal to be proved after those obligations are met.

A Reproducibility, source provenance, and claim ledger

A.1 What is in the archive

The archive contains this PDF and its complete LaTeX source, bibliography, figure-generation source and rendered figures, a native integration patch, the proposed native files, the standalone forest solvers and their separate verifier, focused tests, recorded audit and benchmark output, positive and negative source packets, and integration/reproduction instructions. `README.md` is the entry point; `REPRODUCE.md` gives exact commands. `code_manifest.json` records the proposed code identities, and `SHA256SUMS.txt` records archive file digests.

The native patch is relative to

`8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`.

It is not relative to whichever commit the remote branch later names `main`. The package's patch validation reconstructs the pinned original files, applies the patch, and compares the result byte-for-byte with the supplied proposed files. A fresh complete clone is preferable for the full regression suite because older tests import exact reference implementations and fixtures from historical `reports` directories. Those fixture identities are listed in `historical_fixtures.json` under `results/package_validation/`.

The standalone finite tools require only Python's standard library. The native tests additionally require the pinned ProveIt source tree; Regina is needed for the comparison audit, not for the replay of the seven saved positive packets or three source-authenticated negative packets. Matplotlib is needed only to regenerate the figures. The article uses standard pdfLaTeX packages and BibTeX. Exact commands and recorded versions are included rather than an implicit environment assumption.

A.2 Evidence and theorem mapping

Table 7: How each substantive claim is supported.

Claim	Proof or independent evidence
Native graph root reachability	Lemma 2.3: connected source one-skeleton with positive edges replaced by root stubs.
Geometric reduction of all top handles	Lemma 2.4 and Theorem 2.6; actual belt occurrences, not signed coefficients.
Retention of native exposed ports	Lemma 2.5; a relative boundary-ball argument for authenticated exposed patches.
Polynomial encoded graph stage	Bounds on native chart records and arithmetic in Section 2; no normal-piece expansion.
Stopping characterization	Prior incoming stopping theorem, re-proved for the stated raw update in Section 3.
Complete constrained forest search	Theorem 3.6; independent branch-covering proof-tree replay.
Polynomial sparse-column selection	Reachability and source-SCC lemmas, Rado's criterion, and a proved shortest-path augmentation.
Native total-column bound and parity	Propositions 4.5 and 4.6; triangular face regions and the handle-chain identity.

Claim	Proof or independent evidence
85 to 92 complete native cuts	Pinned baseline/native audit, preserved censuses, 92 independent replays, seven Regina comparisons.
Three interface kernels admit no tree	Saved authenticated negative packets and independent reconstruction of their nondeletable stopping rows.
Finite algorithm agreement	26,995 exhaustive and 7,000 seeded random oracle comparisons; separate positive and negative checkers.
Finite enumeration speed comparison	Specified 2^k-tree family, complete-query timings, raw paired rounds and structural counters.
Native timing crossover	Six supplied-cut sizes and five randomized paired rounds per size; scope and regressions retained.
General quasi-polynomial unknot recognition	Not proved here. Discovery, global-size, pattern-transport, and terminal-completeness obligations remain.

A.3 A small inspectable finite certificate

The following example is the finite form of the two-choice construction:

```
{
  "vertices": 2,
  "columns": 1,
  "edges": [[0, 1], [0, 1]],
  "rows": [[[0, 1]], [[0, 3]]]
}
```

The tree consists of row 1. After deleting it, row 0 is removed on column 0. A positive finite certificate therefore records the tree [1] and the peel pair $[0, 0]$. Changing the selected tree to [0] without changing the source fails verification: the surviving count is three, not one. If both counts are three, the two stopping singleton groups have union of graphic rank one, smaller than their group count two. That rank defect certifies `NO_FOREST`; it does not certify that the associated manifold is not a ball.

The delivered demonstration generates several such positive and negative certificates, replays them with the independent checker, and rejects mutated variants. The full source packets are larger because they also authenticate the triangulation, coordinates, geometry, and interfaces.

A.4 Input validation versus geometric interpretation

A graph/matrix result can be correct as a finite result even when no three-manifold realizes the input. Conversely, a source-authenticated topological result needs the entire geometric chain of implications. The manuscript separates these levels to make review practical:

1. The native source validator establishes the manifold and normal-vector contract and authenticates the complementary charts.
2. The geometric dual theorem identifies valid top cancellations and the retained native port data.
3. The finite matrix theorem decides exactly the stated raw forest-completion problem on those authenticated occurrences.

4. The boundary kernel certifies a genus census when its additional attachment criteria succeed.

The last level can remain inconclusive even when the first three succeed. The ten-core audit demonstrates this distinction directly. A negative finite certificate must not cross these levels as a knot verdict.

There is a second distinction between actual counts and saturated counts. Capping at two is exact for every raw pivot and stopping predicate in this paper. It is not exact for parity: an actual count of three is odd but becomes two. Proposition 4.6 uses authenticated uncapped counts. The source parity audit follows that convention separately from the finite solver's capped matrix representation.

A.5 Attribution and limits of the research claim

The topological cancellation principle is classical. Rado's independent representatives theorem and Edmonds' matroid-intersection framework are classical. Compressed normal-surface cutting is established in the Agol–Hass–Thurston and Lackenby literature. The stopping-set criterion and its raw-cancellation extension already occur in the inspected incoming research. The contribution claimed here is their specific connection to the native source geometry, the retained interface proof, the exact constrained tree-selection formulation and sparse-column solution, the native column constraints, and the tested certificate implementation.

We did not perform an exhaustive historical priority search for every combinatorial lemma, and these results have not undergone external peer review or proof-assistant verification. The claims are stated so that they can be checked or refuted independently. The archive supplies the counterexamples, exact inputs, patch, raw timings, and negative evidence needed for that review. Numerical experiments support implementation correctness and practical scope; they do not replace the proofs or imply the unresolved global asymptotic bound.

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